Experiment 4

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Contents

1 Title

Study of Boolean algebra using AND, OR, NAND, NOR etc. ICs.

2 Aim

In this experiment, we aim to verify the truth tables of basic Boolean logic gates (AND, OR, NOT, NAND, NOR, and XOR) using standard IC chips. Once we confirm how the individual gates behave, we will use them to wire up and test three different combinational logic circuits.

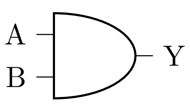
3 Theory

Digital circuits operate differently from analog ones. Instead of processing a continuous range of voltages, they use two distinct voltage levels to represent information. In our lab, a voltage near +5V acts as a 'High' state (Boolean '1'), while a voltage near 0V (ground) is our 'Low' state (Boolean '0').

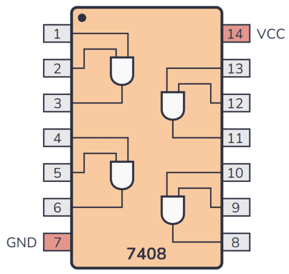
This two-state system is what makes digital logic so reliable against electrical noise. If the voltage fluctuates a little, the gate still recognizes it as a high or a low. To understand how a logic gate or a whole circuit behaves, we map out every possible input combination and the resulting output in a Truth Table.

3.1 2-input AND gate

The AND gate (IC 7408) outputs a '1' only when both of its inputs are tied to a '1'. If either input is grounded (0), the output stays at 0. The mathematical expression is $Y = A \cdot B$.



(a) Logic Symbol



(b) Pinout diagram of AND IC

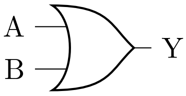
A	B	$Y = AB$
0	0	0
0	1	0
1	0	0
1	1	1

(c) Truth Table

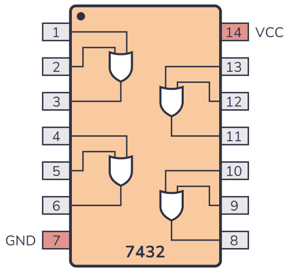
Figure 1: AND Gate Characteristics

3.2 2-input OR gate

The OR gate (IC 7432) gives a '1' output as long as at least one of its inputs is '1'. You only get a '0' if both inputs are 0. The Boolean equation is $Y = A + B$.



(a) Logic Symbol



(b) Pinout diagram of OR IC

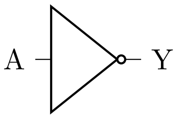
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

(c) Truth Table

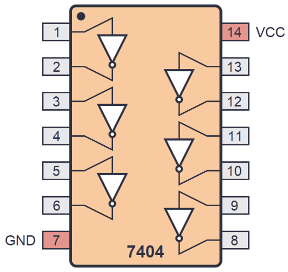
Figure 2: OR Gate Characteristics

3.3 NOT gate

The NOT gate (IC 7404), often called an inverter, simply reverses the input. A '1' becomes a '0', and a '0' becomes a '1'. We write this as $Y = \overline{A}$.



(a) Logic Symbol



(b) Pinout diagram of NOT IC

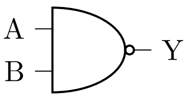
A	$Y = \overline{A}$
0	1
1	0

(c) Truth Table

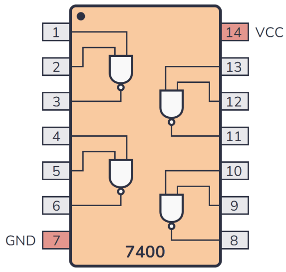
Figure 3: NOT Gate Characteristics

3.4 2-input NAND gate

The NAND gate (IC 7400) works like an AND gate with an inverter attached to the end. It gives a '0' only when both inputs are '1'; otherwise, the output is '1'. The equation is $Y = \overline{A \cdot B}$.



(a) Logic Symbol



(b) Pinout diagram of NAND IC

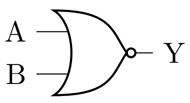
A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

(c) Truth Table

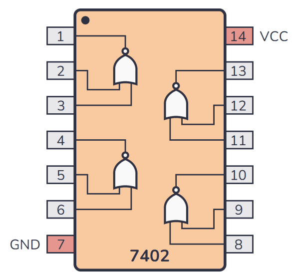
Figure 4: NAND Gate Characteristics

3.5 2-input NOR gate

The NOR gate (IC 7402) is a combination of an OR gate and a NOT gate. It outputs a '1' strictly when both inputs are '0'. Any other combination results in a '0'. The expression is $Y = \overline{A + B}$.



(a) Logic Symbol



(b) Pinout diagram of NOR IC

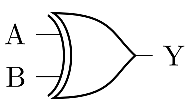
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

(c) Truth Table

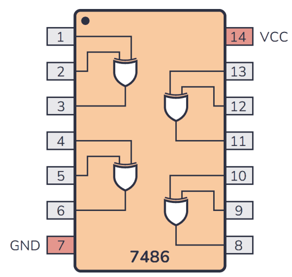
Figure 5: NOR Gate Characteristics

3.6 2-input XOR gate

The Exclusive-OR gate (IC 7486) acts like a difference detector. It outputs a '1' if the two inputs are different from each other. If both inputs match (both 0s or both 1s), the output is 0. The equation is $Y = A \oplus B$.



(a) Logic Symbol



(b) Pinout diagram of XOR IC

A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

(c) Truth Table

Figure 6: XOR Gate Characteristics

4 Circuit Diagrams

After testing the gates individually, we wired up the following three combinational logic circuits on the breadboard to observe how the logic cascades from one gate to the next.

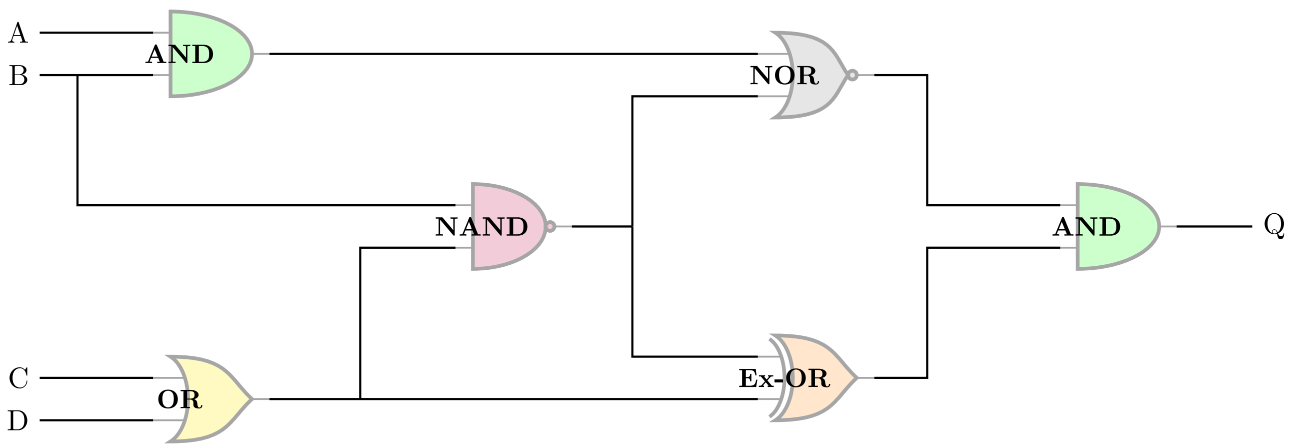


Figure 7: Logic Circuit 1 Diagram

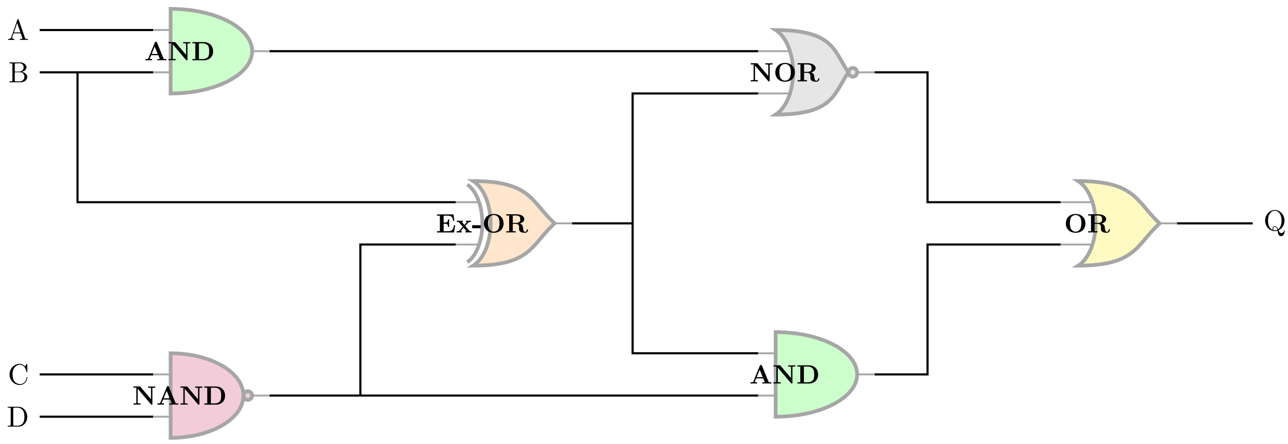


Figure 8: Logic Circuit 2 Diagram

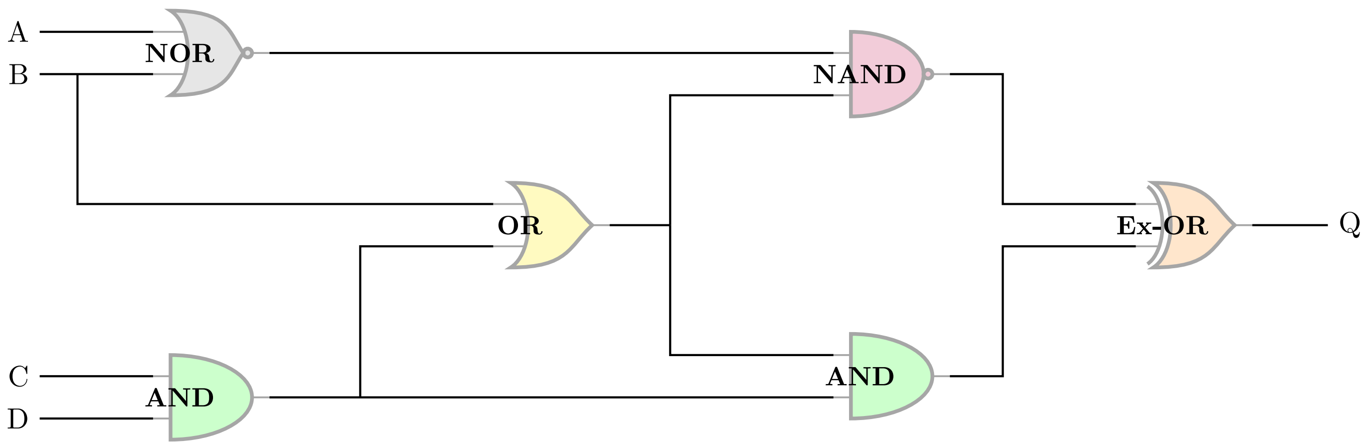


Figure 9: Logic Circuit 3 Diagram

5 Data

Table 1: Experimental Observations for Logic Circuits 1, 2, and 3

Sl. No.	A	B	C	D	Example 1 Q	Example 2 Q	Example 3 Q
1	0	0	0	0	0	1	1
2	0	0	0	1	0	1	1
3	0	0	1	0	0	1	1
4	0	0	1	1	0	1	1
5	0	1	0	0	0	1	1
6	0	1	0	1	1	1	1
7	0	1	1	0	1	1	1
8	0	1	1	1	1	0	0
9	1	0	0	0	0	1	1
10	1	0	0	1	0	1	1
11	1	0	1	0	0	1	1
12	1	0	1	1	0	1	0
13	1	1	0	0	0	0	1
14	1	1	0	1	0	0	1
15	1	1	1	0	0	0	1
16	1	1	1	1	0	0	0

6 Conclusion

The experiment went quite well. We successfully verified the basic truth tables for all the individual logic gate ICs without any issues. Beyond that, the three combinational logic circuits we mapped out on the breadboard behaved exactly as we expected mathematically. The experimental outputs matched our theoretical truth tables nicely, confirming the logic flows.